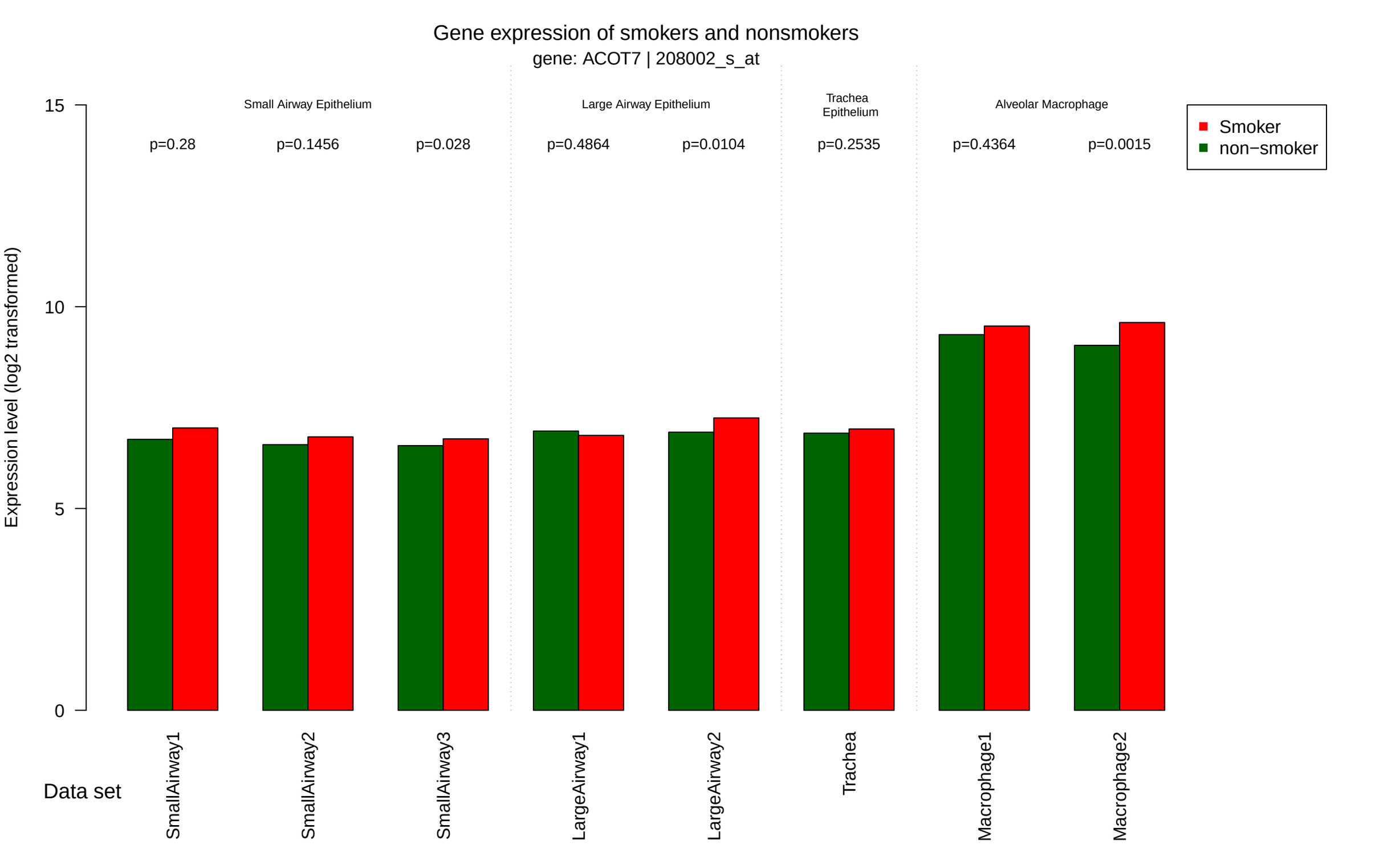
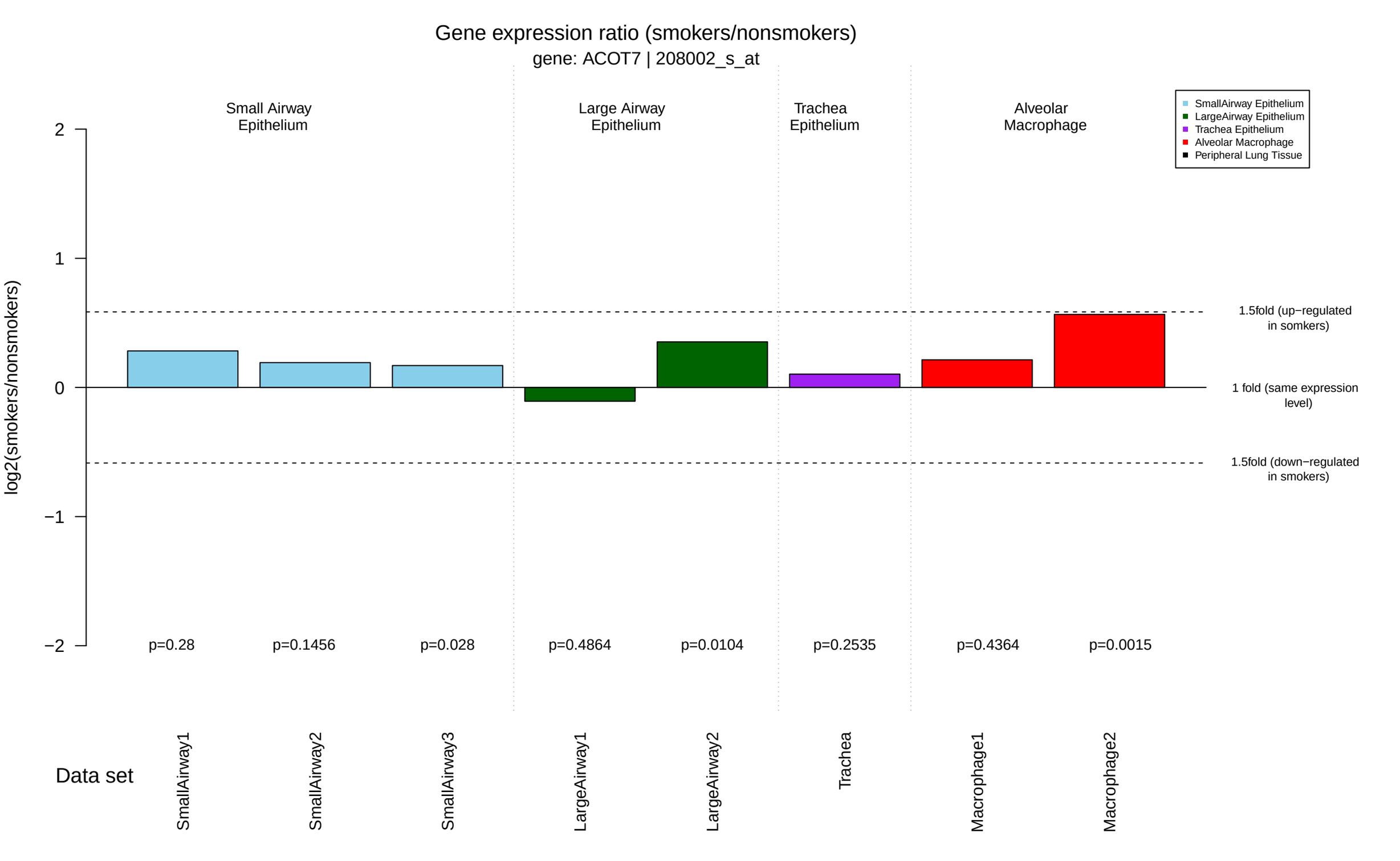
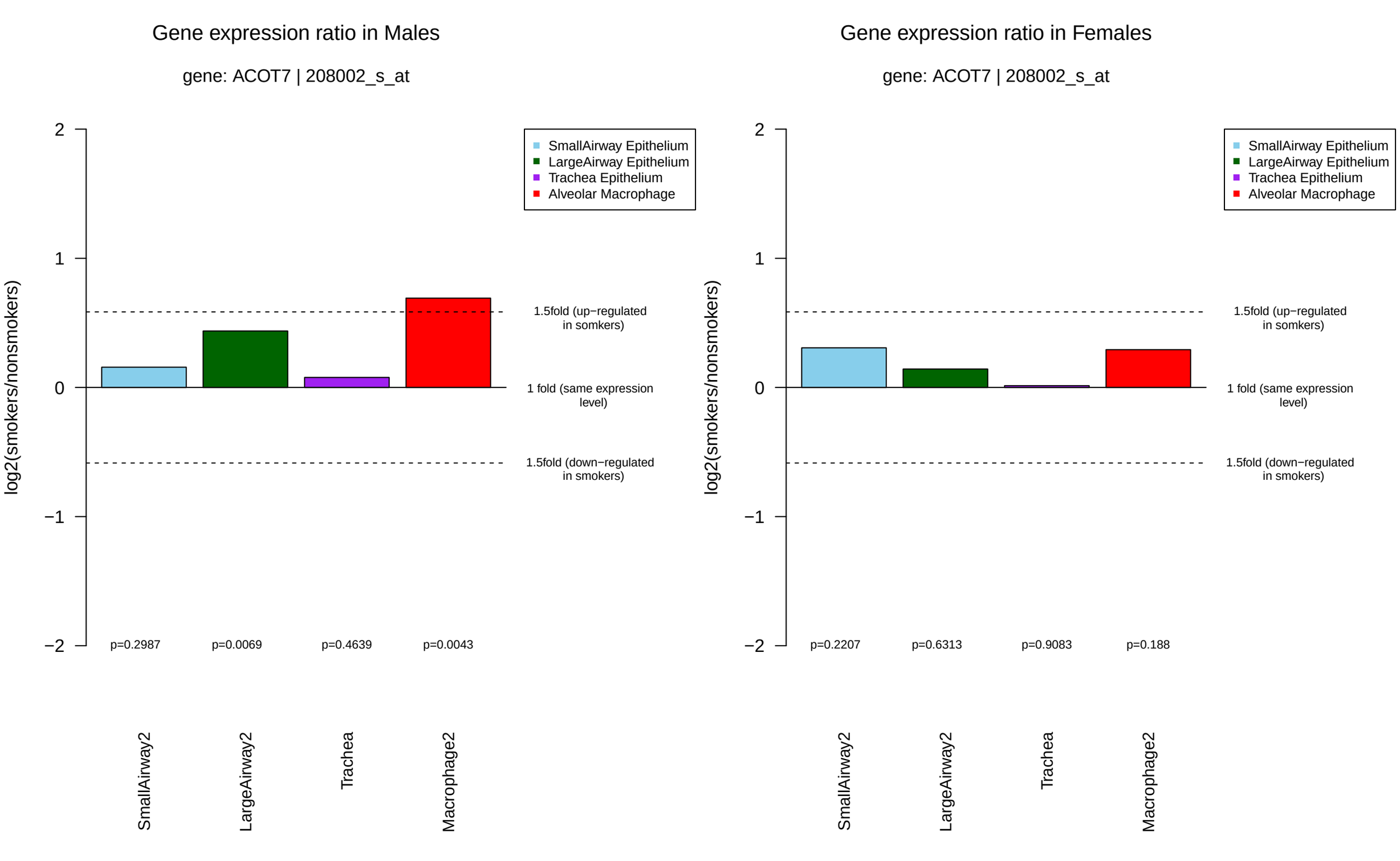
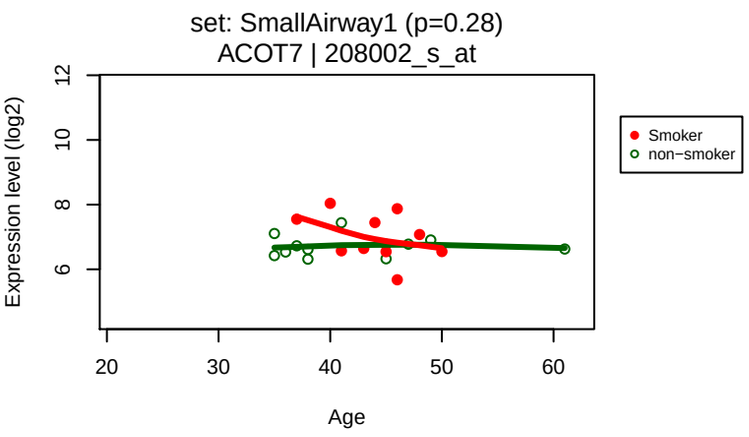
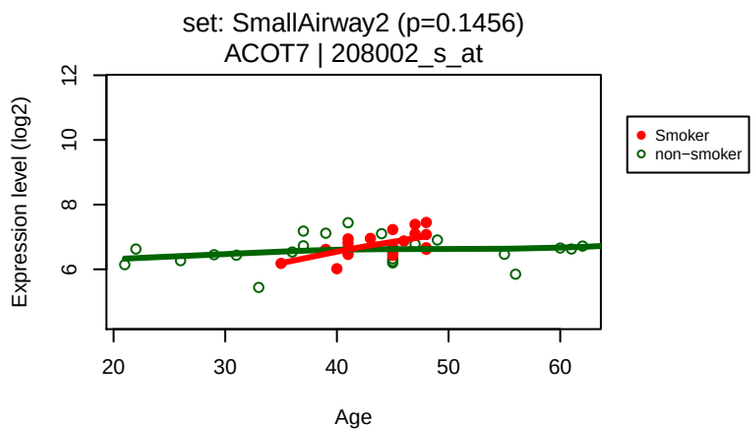
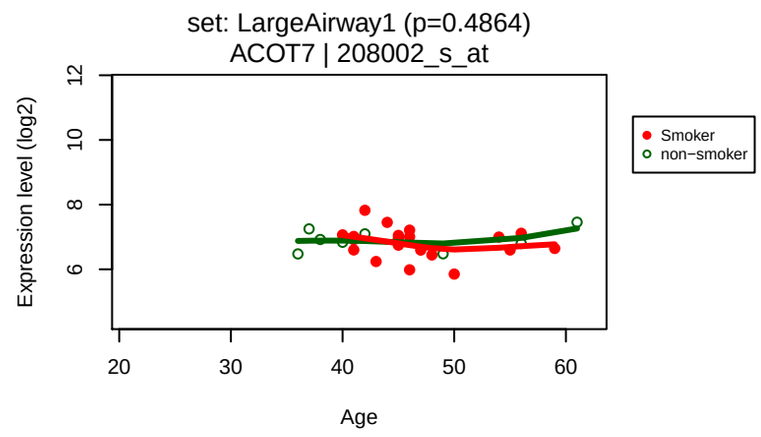

Gene expression of smokers and nonsmokers
gene: ACOT7 | 208002_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.28
p=0.1456
p=0.028
p=0.4864
p=0.0104
p=0.2535
p=0.4364
p=0.0015
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

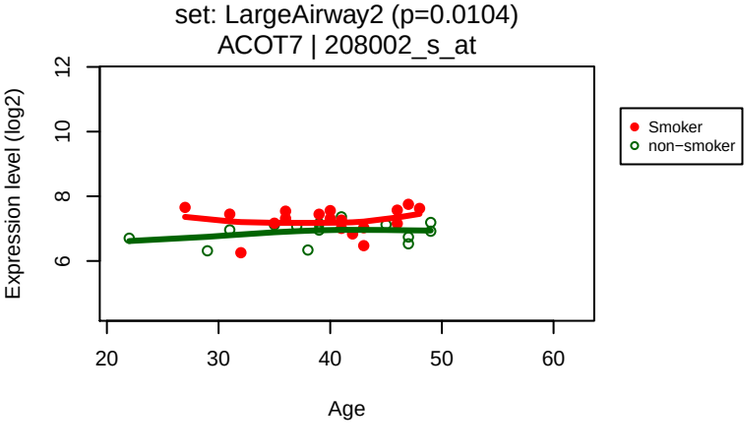

Gene expression ratio (smokers/nonsmokers)
gene: ACOT7 | 208002_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.28
p=0.1456
p=0.028
p=0.4864
p=0.0104
p=0.2535
p=0.4364
p=0.0015
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

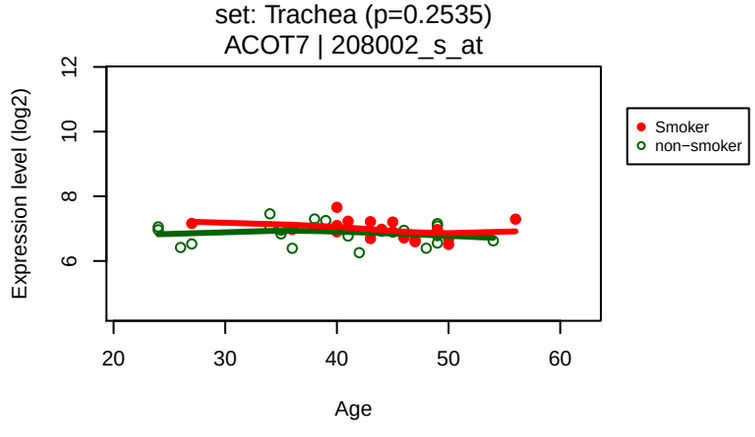

Gene expression ratio in Males
gene: ACOT7 | 208002_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2987
p=0.0069
p=0.4639
p=0.0043
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACOT7 | 208002_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2207
p=0.6313
p=0.9083
p=0.188
SmallAirway2
LargeAirway2
Trachea
Macrophage2

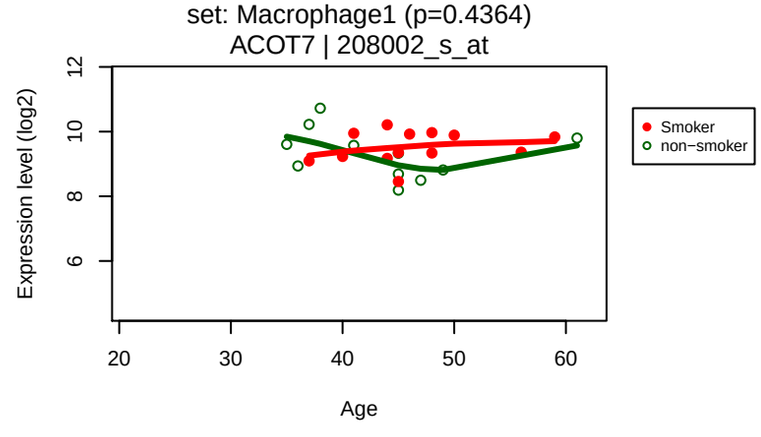

set: SmallAirway1 (p=0.28)
ACOT7 | 208002_s_at
Expression level (log2)
Age
Smoker
non−smoker

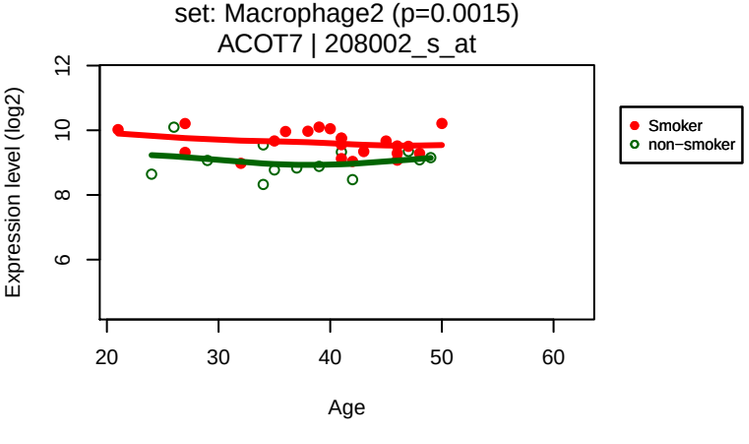

set: SmallAirway2 (p=0.1456)
ACOT7 | 208002_s_at
Expression level (log2)
Age
Smoker
non−smoker

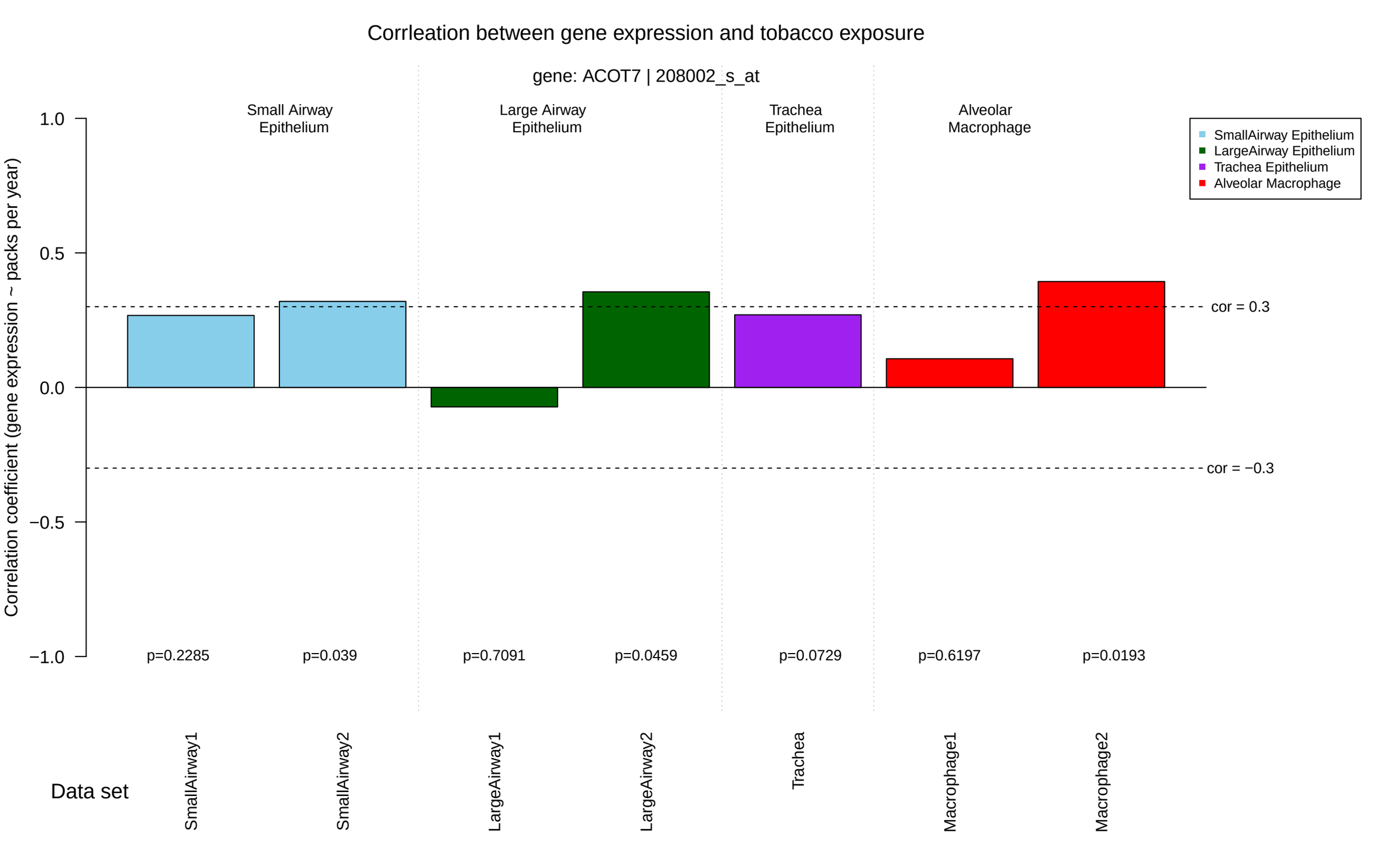

set: LargeAirway1 (p=0.4864)
ACOT7 | 208002_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0104)
ACOT7 | 208002_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2535)
ACOT7 | 208002_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4364)
ACOT7 | 208002_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0015)
ACOT7 | 208002_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACOT7 | 208002_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2285
p=0.039
p=0.7091
p=0.0459
p=0.0729
p=0.6197
p=0.0193
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2